

Ismail AISSAOUI

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To the Recruitment Committee
University of Geneva (UNIGE)
GSEM - Research Institute for Statistics (RISIS)
Attn: Professor Sebastian Engelke

May 4, 2026

**Subject: Application for PhD Candidate / Research and Teaching Assistant position
(Ref: 6889)**

Dear Professor Engelke,

I am writing to formally submit my application for the PhD Candidate and Teaching Assistant position at the RISIS Institute, University of Geneva. As a State Engineer in Artificial Intelligence from ESI-SBA, I am eager to join your research group to conduct statistical analysis of AI weather forecasting models at the intersection of Extreme Value Theory and Climate Sciences.

My application is supported by a personal recommendation from **Prof. Vincent Cheutet (INSA Lyon)**, who identified my background in Graph Neural Networks (GNNs) and hybrid modeling as a strong match for your group's focus on complex AI architectures like **GraphCast** and **AIFS**.

Currently, for my thesis at Sonatrach, I am developing **Physics-Informed Digital Twins** for the monitoring of industrial systems. This work has allowed me to master the mathematical and practical aspects of several domains directly relevant to your research:

- **Graph-based AI:** I have extensive experience with GNNs, which are the backbone of GraphCast. I am particularly interested in how the message-passing mechanisms on multi-mesh grids can be statistically analyzed to understand their behavior during extreme events.
- **Sequential Dynamics:** I utilize State Space Models (**Mamba-SSM**) and **xLSTM** for trajectory forecasting. My experience with these architectures provides a solid foundation for analyzing the long-range temporal dependencies in climate models.
- **Model Reliability:** My research on "Validity Envelopes" for Digital Twins shares a core objective with your focus on robustness and extremes: ensuring that AI models remain reliable when extrapolating to rare or unseen scenarios.

Beyond research, I am highly motivated by the **Teaching Assistant** aspect of this position. I possess the self-discipline and communication skills required to lead tutorials and supervise Master's students, and I look forward to contributing to the academic life of the GSEM.

I am confident that my technical proficiency in Python and my engineering-grounded perspective on AI will bring a valuable "model-centric" approach to your statistical analysis of climate extremes.

Thank you for considering my application. I look forward to the possibility of discussing my research interests with you.

Sincerely,

Ismail AISSAOUI